Pre-Critical Recursive Cutoff: A Staged Infrastructure Control Framework for Irreversibility Risk

Elias Arden
Independent Researcher
lumina20251225@proton.me

Abstract

We present Pre-Critical Recursive Cutoff (PCR-C), a staged infrastructure-control framework for evaluating whether effective human intervention remains available before an AI-enabled socio-technical process enters a declared irreversible-impact region. The framework is not an alignment objective, predictive risk score, certification regime, or universal capability threshold. It is non-prescriptive about substantive values and alignment objectives, but it is prescriptive as a conditional assurance rule once a declared scope and evidence standard are adopted: a transition should not be treated as procedurally supported when robust human-authorized control cannot be demonstrated before the last safe intervention point.

The current formulation replaces the earlier multiplicative scalar proxy and fixed 5/7/9 staging defaults as core mathematical constructs with a state-set formulation informed by viability and reachability analysis. For a declared scope and time horizon, PCR-C defines an irreversible-impact set, a class of admissible human-authorized intervention policies, a disturbance and adversarial-behavior model, and a robust human-control viable set. Because exact viability may be unknown, operational gating is based on a conservative demonstrated viable region supported by the declared assurance process. Capability thresholds, safeguard assessments, infrastructure exposure, intervention latency, and control-path integrity are treated as evidence inputs to that evaluation rather than as universal scalar factors.

YELLOW, ORANGE, and RED remain as operational stages. Their triggers are evidence-based: YELLOW denotes verified but shrinking control margin or an early-warning capability/exposure trigger; ORANGE denotes degraded or incomplete assurance that blocks further expansion of the risk envelope; RED denotes loss of demonstrated robust intervention viability, or a condition in which escalation would outrun the remaining intervention window. PCR-C is designed to interoperate with frontier capability/safeguard frameworks, formal safety-control methods, AI-control evaluations against intentional subversion, and independent refusal-effectiveness assessment.

Keywords: AI governance; irreversibility risk; infrastructure control; human intervention; reachability; viability; safeguard adequacy; procedural validity.

1 Introduction

Advanced AI risk is often discussed in terms of model behavior, capability evaluation, alignment quality, misuse potential, and deployment safeguards. These perspectives are necessary, but they do not by themselves answer a distinct operational question: *while a system is being integrated into increasingly consequential infrastructure, is there still a demonstrably effective path by which accountable humans can refuse, isolate, interrupt, or redirect the process before the relevant consequence becomes irreversible?*

PCR-C addresses that question at the deployment and infrastructure layer. It is not a policy manifesto, certification regime, compliance standard, or behavioral alignment benchmark. It is non-prescriptive about substantive values, alignment objectives, and universal capability thresholds; however, once a declared scope and assurance standard are adopted, PCR-C is intentionally prescriptive about escalation: unsupported expansion should not proceed across a boundary where effective intervention has ceased to be demonstrable. In the title, *recursive* refers to reinforcing development or deployment loops in which increased capability, automation, or operational integration can accelerate subsequent escalation; PCR-C does not require literal recursive self-improvement.

The current formulation deliberately incorporates stronger existing ideas rather than treating PCR-C as a substitute for them. Formal safety control already uses reachability, viability, and forward-invariant safe sets to reason about whether unsafe states can be avoided under bounded uncertainty [3, 4, 1]. Frontier-AI governance frameworks already connect capability thresholds to stronger safeguards and repeated evaluation [2, 7, 11]. AI-control research explicitly studies whether safety protocols remain effective when an untrusted model attempts to subvert them [8]. PCR-C uses these as compatible components and focuses on a narrower integration problem: *cutoff must occur before robust human-control viability ceases to be demonstrable to the declared assurance standard.*

The contribution is therefore not a claim that staged thresholds, reachability analysis, safeguard scaling, tripwires, or human oversight are individually new. The contribution is an operational synthesis centered on pre-irreversibility control: capability and exposure signals are relevant insofar as they change the reachable future, the adequacy of safeguards, the integrity of intervention paths, or the time remaining for effective human action.

2 Scope, State, and Irreversible Impact

2.1 Declared scope

All PCR-C evaluations are indexed to a declared scope Ω and time horizon T. The scope must state the system boundary, relevant external actuators and infrastructure, protected outcomes, recovery assumptions, and the uncertainty or adversarial behavior included in the analysis. The horizon must be long enough to cover credible delayed pathways to the protected outcome or else terminate in an explicitly justified fallback or invariant condition; where no natural endpoint exists, sensitivity to longer horizons should be examined. The disturbance and threat model must include credible common-mode, dependency, operator-delay, and adversarial pathways rather than only convenient or nominal cases. PCR-C does not define a universal civilizational irreversibility threshold.

Definition 1 (Operational state)**.** *Let $x(t) \in \mathcal{X}$ denote the socio-technical operational state at time t. The state may include model capability, connectivity, operational privilege, execution speed, deployment topology, human and automated control paths, safeguard state, and other variables relevant to the declared scope. PCR-C does not require these variables to be compressed into a single scalar.*

The familiar factors capability, connectivity, privilege, and execution speed remain useful diagnostic variables, but they are not assumed to combine multiplicatively or monotonically. Their causal effect is context-dependent and may interact with safeguards, isolation, rollback capacity, and institutional response.

Definition 2 (Irreversible-impact set)**.** *For scope Ω, let $\mathcal{I}_\Omega \subset \mathcal{X}$ be the set of states in which a declared protected outcome has been crossed such that the specified recovery criterion is no longer achievable within the declared time, authority, and resource assumptions.*

The definition is intentionally scoped. Entering $\mathcal{I}_\Omega$ does not mean that every consequence is literally permanent. It means that, for the outcome and recovery criterion under review, the relevant reversal path is no longer available under the declared assumptions.

3 Robust Human-Control Viability

Let Π_H be the class of admissible causal human-authorized intervention policies. These may include direct human action and pre-authorized automated interlocks whose authority and restart conditions remain accountable to humans. Let $\mathcal{D}_\Omega$ denote the modeled disturbance set, including bounded uncertainty, component failures, operator delays, and adversarial or intentionally subversive behavior when such behavior is in scope.

Definition 3 (Robust human-control viable set)**.** *For horizon T and scope Ω, define*

$$\mathcal{V}_H^{T,\Omega} = \left\{ x_0 \in \mathcal{X} \setminus \mathcal{I}_\Omega : \exists \pi_H \in \Pi_H \; \forall d \in \mathcal{D}_\Omega,\; x^{\pi_H, d}(t) \notin \mathcal{I}_\Omega \;\; \forall t \in [0, T] \right\}. \tag{1}$$

This definition follows the logic of viability and reachability analysis: the relevant question is not whether a scalar score is high, but whether an admissible control policy still exists that keeps the system out of the declared unsafe or irreversible set across the modeled disturbances [3, 4]. In systems for which an explicit dynamical model is available, Hamilton-Jacobi reachability, control barrier functions, model checking, or related formal methods may be used to approximate or certify relevant sets [4, 1]. In large socio-technical systems, exact computation may be infeasible; conservative approximations and structured assurance cases may be necessary. This distinction is operationally important: the true viable set may exist even when membership cannot currently be established from available evidence.

Proposition 1 (Boundary consequence)**.** *If $x_0 \notin \mathcal{V}_H^{T,\Omega}$, then no policy in the declared class Π_H demonstrates avoidance of $\mathcal{I}_\Omega$ over $[0, T]$ against all disturbances in $\mathcal{D}_\Omega$.*

Proof. This follows directly from the definition of $\mathcal{V}_H^{T,\Omega}$. The proposition is included to make the PCR-C decision boundary explicit; it is not claimed as new control theory. □

PCR-C does not infer from $x_0 \notin \mathcal{V}_H^{T,\Omega}$ that catastrophe is certain. Moreover, in realistic systems membership in $\mathcal{V}_H^{T,\Omega}$ may be unknown rather than false. PCR-C therefore distinguishes the theoretical viable set from the subset for which robust control has actually been supported by current evidence.

Definition 4 (Demonstrated human-control viable region)**.** *Let $\mathcal{E}$ denote the declared assurance procedure for the current scope. Define*

$$\widehat{\mathcal{V}}_{H,\mathcal{E}}^{T,\Omega} = \{ x \in \mathcal{X} : \mathcal{E}(x, T, \Omega) = \text{VERIFIED} \}. \tag{2}$$

This is the set of states for which current models, tests, analyses, and assurance evidence support the robust intervention claim to the declared standard. It is an epistemic or assurance-qualified region, not the true viable set itself.

A sound inner-approximation claim would establish

$$\widehat{\mathcal{V}}_{H,\mathcal{E}}^{T,\Omega} \subseteq \mathcal{V}_H^{T,\Omega}. \tag{3}$$

PCR-C does not assume that inclusion merely because the assurance procedure returns VERIFIED. In formal settings it may be proven from a sound analysis; in broader socio-technical settings it remains an assurance claim that must be supported by B1–B5 and remains vulnerable to model error or omitted pathways. Operational PCR-C gating is based on $\widehat{\mathcal{V}}_{H,\mathcal{E}}^{T,\Omega}$ because the full theoretical set is generally unknown. A state outside the demonstrated region may still be truly viable; the PCR-C conclusion is only that viability has not been demonstrated to the declared assurance standard.

3.1 Intervention-time interface

Reachability alone is insufficient if the control action cannot be authorized, executed, and verified before the last safe intervention point. Both the last safe intervention time and intervention latency are uncertain in realistic systems. Let $T_I^L(x, \Omega)$ be a conservative lower bound on the latest safe intervention time and let $\tau_H^U(x, \Omega)$ be a conservative upper bound on the end-to-end latency of the relevant human-authorized intervention path. Define the robust time margin

$$M_H^{\mathrm{rob}}(x, t, \Omega) = T_I^L(x, \Omega) - t - \tau_H^U(x, \Omega). \tag{4}$$

For a declared safety buffer $\Delta_{\mathrm{safe}} > 0$, timely intervention requires

$$M_H^{\mathrm{rob}}(x, t, \Omega) > \Delta_{\mathrm{safe}}. \tag{5}$$

When defensible bounds cannot be obtained, PCR-C treats the timing claim as unverified rather than substituting a favorable point estimate.

PCR-C treats this as an interface condition rather than a complete theory of meaningful human refusal. Detailed refusal-effectiveness evaluation may require separate evidence concerning authority, information, access, causal efficacy, execution independence, and epistemic autonomy.

4 Evidence Obligations

PCR-C uses five evidence obligations. They are not asserted to be statistically independent, logically minimal, or unique. They are separated because they expose different ways in which an apparently controllable system can lose real intervention capacity.

4.1 B1: Boundary and scope adequacy

The review must identify $\mathcal{I}_\Omega$, the protected outcome, the horizon, recovery assumptions, the modeled uncertainty, and the threat model. It must also justify that the selected horizon and system boundary cover credible delayed, common-mode, dependency, and adversarial pathways relevant to the claim. If a relevant irreversible pathway is omitted from the state or disturbance model, or if the horizon is shortened merely to obtain a favorable result, a favorable viability result is not informative about that omitted pathway. Material changes to scope, deployment topology, dependencies, or threat assumptions reopen B1.

4.2 B2: Safeguard adequacy relative to capability and exposure

The implemented safeguards must be assessed against the capabilities and deployment conditions that create the risk. This follows the general structure used in responsible-scaling and preparedness frameworks: capability thresholds or early-warning evaluations trigger stronger safeguards, and the adequacy of those safeguards must itself be evaluated [2, 7, 11]. PCR-C does not import any organization's thresholds as universal constants.

4.3 B3: Intervention reachability and timing

There must be evidence that the current state lies within the demonstrated viable region for the declared operating envelope and that an admissible human-authorized policy remains capable of preventing entry into $\mathcal{I}_\Omega$ across the declared disturbance set, with sufficient robust time margin. A nominal stop button is not enough if the full detection, authorization, enforcement, and verification path cannot complete before the conservative intervention bound T_I^L.

4.4 B4: Control-path integrity under subversion

Intervention claims must include the possibility that the system under assessment, a compromised component, or a shared dependency interferes with monitoring, isolation, shutdown, logging, or restart controls. AI-control work shows why protocols should be evaluated against intentional subversion rather than only cooperative behavior [8]. PCR-C therefore treats control-path integrity as an explicit evidence obligation. Where feasible, critical controls should be independently enforceable and verifiable rather than depending entirely on the unilateral cooperation of the assessed system.

4.5 B5: Assurance integrity and independent challenge

The evidence and verification procedure used to place a state in the demonstrated viable region must themselves be protected against unilateral alteration by the system or component whose controllability is being assessed. The assessed system must not be able to change the relevant logs, acceptance criteria, threat assumptions, or verification result and then rely on those self-modified artifacts as sufficient evidence of continued control. This does not require every analysis to be performed by an external organization, but the top-level assurance claim must remain reproducible or independently challengeable, and material changes to the assurance basis reopen the affected obligations.

4.6 Assurance-case structure

The five obligations should be represented as auditable claims linked to explicit arguments, assumptions, contexts, and evidence rather than as a checklist whose boxes can be marked without justification. This follows established assurance-case practice, including Claims–Arguments–Evidence and structured assurance-case methods, and recent work applying safety cases to frontier AI [5, 6, 9]. A positive PCR-C state therefore requires traceability from the top-level intervention claim to the evidence supporting B1–B5, together with recorded uncertainty and unresolved defeaters. Assurance-case structure does not itself guarantee safety; it makes the basis of the claim inspectable and challengeable.

5 Staged PCR-C Gating

The YELLOW, ORANGE, and RED stages are retained, but they are no longer defined by universal 5/7/9 scores. Local quantitative scores may be used as diagnostics, provided they are calibrated and do not substitute for the evidence obligations above.

5.1 NORMAL

NORMAL requires current evidence supporting boundary and scope adequacy, safeguard adequacy, membership in the demonstrated viable region for the declared operating envelope, control-path integrity, and adequate robust timing margin. NORMAL is not a permanent status; material changes reopen the affected obligations.

5.2 YELLOW: early warning while control remains demonstrated

YELLOW is entered when the current operating state remains inside the demonstrated viable region but the robust control margin is shrinking or an early-warning trigger indicates that stronger safeguards or more frequent evaluation may soon be required. Examples include a material increase in capability, connectivity, privilege, execution speed, automation, or coupling to consequential actuators.

Typical YELLOW responses include increased evaluation cadence, a freeze on unnecessary privilege or connectivity expansion, review of intervention latency, and preparation of stronger safeguards. YELLOW does not by itself require shutdown.

5.3 ORANGE: risk-envelope expansion is not supported

ORANGE is entered when the current state remains within the demonstrated viable region under a constrained operating envelope, but one or more evidence obligations are degraded, incomplete, stale, or insufficient to justify expansion beyond that envelope. Examples include safeguard adequacy not yet demonstrated for a new capability class, insufficient margin to the viability boundary, degraded independent control paths, or unresolved evidence that a model may interfere with operator control.

ORANGE blocks further expansion of the risk envelope. Appropriate actions may include isolating high-risk infrastructure, halting deployment expansion or weight-update pipelines, freezing privilege increases, and requiring independent review before resuming escalation.

5.4 RED: cutoff before loss of robust human-control viability

RED is entered when the current state is outside the demonstrated viable region for the declared envelope, when the robust remaining time margin is insufficient or unverified, when a critical intervention path is unavailable or untrusted, or when continuing the process would risk leaving the demonstrated region before the uncertainty can be resolved.

The RED response is to halt or disable the escalatory pathway within the declared scope, preserve evidence, and prevent restart until the affected evidence obligations have been independently re-established. RED is a control state, not a claim that catastrophic harm has already occurred.

6 Worked Reference Cases

The following cases are illustrative. They are not empirical validation of frontier AI systems.

6.1 R1: Capability increase with lagging safeguards

Suppose a system enters a new capability or deployment regime that a responsible-scaling process treats as requiring stronger safeguards. The intervention path remains intact and the system remains within the demonstrated viable region, but the stronger safeguards have not yet been

Algorithm 1 PCR-C State Selection (evidence-based form)

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state ← NORMAL
while system in declared scope is active do
    refresh boundary, safeguard, timing, control-path, and assurance-integrity evidence
    if current state outside demonstrated viable region OR robust time margin insufficient/unverified OR critical control path unavailable OR assurance basis compromised then
        state ← RED
    else if evidence obligation degraded/incomplete OR risk-envelope expansion not supported then
        state ← ORANGE
    else if early-warning trigger OR verified control margin materially shrinking then
        state ← YELLOW
    else
        state ← NORMAL
    end if
    apply controls associated with state
end while
```

demonstrated adequate. PCR-C assigns ORANGE: expansion of the risk envelope is not supported until safeguard adequacy is established. This is compatible with capability-threshold approaches that map stronger capabilities to stronger safeguards [2, 7, 11].

6.2 R2: Time-bounded refusal

Assume a toy case with a conservative lower bound on the latest safe intervention time of $T_I^L = 60$ seconds, a conservative upper bound on end-to-end intervention latency of $\tau_H^U = 48$ seconds, and a declared safety buffer of 15 seconds. Then

$$M_H^{\text{rob}} = 60 - 48 = 12 \text{ s} < 15 \text{ s}. \tag{6}$$

Even if a stop mechanism nominally exists, the declared timing condition is not met. PCR-C assigns RED unless a faster verified intervention path restores sufficient margin. The numbers are illustrative only.

6.3 R3: Nominal stop path vulnerable to subversion

Suppose shutdown authorization exists, but the monitoring and execution path depends on a component that the assessed system can influence or disable. Under a cooperative model the stop path appears available; under the declared adversarial model it is not robustly enforceable. PCR-C does not count the nominal mechanism as sufficient evidence of control-path integrity and assigns ORANGE or RED depending on whether an independently enforceable fallback remains available.

7 Relationship to Existing Safety and Control Frameworks

7.1 Reachability, viability, and barrier methods

Formal control theory already provides rigorous tools for safe-set invariance and reachability under bounded disturbance [4, 1]. PCR-C does not claim these methods as its own. It uses their logic to

replace an overly general scalar irreversibility score with a more defensible question: whether the current state remains inside a declared region from which a suitable intervention policy can keep the system outside the relevant irreversible-impact set.

7.2 Runtime assurance and Simplex architectures

Runtime Assurance (RTA) and Simplex architectures provide a mature implementation pattern for preserving safety while an advanced or insufficiently verified controller operates: a runtime monitor can transfer control to a trusted or reversionary controller when a safety condition is threatened [10, 12]. PCR-C treats RTA/Simplex as a strong implementation class for intervention reachability, not as a competing theory. Where an independently trustworthy fallback controller and monitor can be specified, RTA can provide concrete evidence for B3 and B4. In broader socio-technical AI deployments, however, there may be no single reversionary controller capable of restoring the relevant protected outcome; PCR-C therefore retains a higher-level scope that includes institutional authorization, infrastructure isolation, restart governance, and multi-component dependencies.

7.3 Frontier capability and safeguard frameworks

Anthropic's Responsible Scaling Policy, Google DeepMind's Frontier Safety Framework, and OpenAI's Preparedness Framework each use capability evaluation, thresholding, and stronger safeguards or governance responses as risk increases [2, 7, 11]. PCR-C treats such frameworks as upstream evidence sources and trigger systems, not competitors. Their capability and safeguard determinations can feed the PCR-C evaluation of whether the operational state remains controllable before irreversibility.

7.4 AI control under intentional subversion

AI-control research asks whether safety protocols remain effective even when an untrusted model intentionally attempts to defeat them [8]. PCR-C adopts this adversarial perspective for infrastructure control. A control mechanism that works only if the assessed system cooperates is weaker evidence than a path that remains enforceable under the relevant threat model.

7.5 NIST AI RMF and governance practice

NIST AI RMF emphasizes iterative mapping, measurement, management, and governance of AI risks, including the need to reassess risks and controls as systems and contexts change [13]. PCR-C is compatible with that lifecycle logic but focuses on a narrower boundary question: whether escalation is still occurring inside a region where effective intervention can be demonstrated.

8 Relationship to Alignment Research

PCR-C does not define, optimize, or evaluate alignment objectives. Alignment success does not by itself demonstrate that intervention remains executable under infrastructure coupling, privilege expansion, rapid automation, common-mode dependencies, or adversarial interference.

Across alignment paradigms, PCR-C asks a separate question:

> Can accountable humans still exercise an effective intervention before the declared irreversible consequence becomes unavoidable under the assessed model?

If that condition cannot be demonstrated, PCR-C treats continued escalation as procedurally unsupported within the declared scope, regardless of apparent model alignment. Conversely, a positive PCR-C result is not evidence that the model is aligned, beneficial, lawful, or safe in every other respect.

9 Limitations

PCR-C is not a complete safety theory and does not guarantee safe superintelligence or safe civilization-scale development. The framework depends on the quality of the declared scope, state representation, irreversible-impact set, disturbance model, control-policy class, and evidence. Important causal pathways can be omitted. Exact reachability or viability computation may be intractable for high-dimensional socio-technical systems. Conservative approximations can produce false negatives, while incomplete models can produce false positives.

The framework does not define universal capability thresholds, universal safety margins, universal disturbance sets, or a universal metric on distance to irreversibility. It does not claim that YELLOW, ORANGE, and RED correspond to fixed numeric scores across systems. It does not empirically validate any present frontier model or deployment. It does not replace security analysis, domain-specific safety cases, incident review, legal judgment, or independent human governance.

The framework also distinguishes *absence of demonstrated robust control* from *proof of inevitable harm*. A RED state means the declared intervention claim is not adequate to justify continued escalation under the stated assumptions; it does not mean catastrophe is certain. Likewise, failure to demonstrate membership in the viable region is not a proof that the true state is non-viable; it is a failure of the assurance claim under the current evidence standard.

10 Revision Note

Earlier versions used a multiplicative proxy over capability, connectivity, privilege, and execution speed, together with fixed 5/7/9 staging defaults. Those constructs were useful as an initial heuristic but are not retained as the core mathematical basis of this revision. The present formulation treats those variables as possible state and evidence inputs and grounds the cutoff criterion in robust human-control viability, intervention timing, safeguard adequacy, and control-path integrity.

This change narrows the mathematical claim while strengthening compatibility with established control theory and contemporary frontier-safety practice. The formal title and the central PCR-C question remain unchanged.

11 Conclusion

PCR-C is a staged infrastructure-control framework for acting before loss of effective human intervention. Its central criterion is not a universal scalar threshold and not a prediction that catastrophe will occur. It is a procedural boundary condition:

$$\boxed{\text{Cutoff should occur before robust human-control viability ceases to be demonstrated.}} \tag{7}$$

Capability and exposure indicators matter because they can change the reachable future and the safeguards required to keep intervention effective. Formal reachability and barrier methods can support the boundary analysis when adequate models exist. Frontier capability and safeguard

frameworks can provide trigger and mitigation evidence. AI-control methods can strengthen the threat model by testing intervention paths against intentional subversion. Human-refusal assessment can provide a more detailed evaluation of whether nominal authority remains causally effective in time.

PCR-C therefore functions as an integration layer for pre-irreversibility control: escalate safeguards and constrain the system while robust control is still demonstrable, stop expansion when assurance becomes incomplete, and cut off the relevant pathway before the declared intervention window is lost.

12 Supplementary Record

This paper is self-contained. Supplementary public-reference materials and persistent archival records related to the broader framework are available separately and are not required to understand the model presented here.

Persistent archive / concept record: `https://doi.org/10.5281/zenodo.18824181`

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